OS ORANGE ASSIGNMENT

SRN: PES2UG24CS124

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Github link: https://github.com/Boffin-Omega/PES2UG24CS124-pes-vcs

Screenshots:

Phase 1

1 A) Output of `./test_objects`

```
gagangowda@DESKTOP-MBRTPOS:~/PES2UG24CS124-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1 B) Output of `find .pes/objects -type f`

```
gagangowda@DESKTOP-MBRTPOS:~/PES2UG24CS124-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
```

Phase 2

2 A) Output of `./test_tree`

```
gagangowda@DESKTOP-MBRTPOS:~/PES2UG24CS124-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2 B) Raw binary of tree object

```
gagangowda@DESKTOP-MBRTPOS:~/PES2UG24CS124-pes-vcs$ xxd .pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 | head
-20
00000000: 626c 6f62 2036 0068 656c 6c6f 0a         blob 6.hello.
```

Phase 3

3 A) pes init -> pes add -> pes status

```
gagangowda@DESKTOP-MBRTPOS:~/PES2UG24CS124-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
gagangowda@DESKTOP-MBRTPOS:~/PES2UG24CS124-pes-vcs$ ./pes add file1.txt file2.txt
gagangowda@DESKTOP-MBRTPOS:~/PES2UG24CS124-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  test_objects
  untracked:  test_objects.c
  untracked:  Makefile
  untracked:  test_tree
  untracked:  object.c
  untracked:  index.c
  untracked:  README.md
  untracked:  tree.h
  untracked:  test_sequence.sh
  untracked:  pes.h
  untracked:  index.h
  untracked:  pes.c
  untracked:  commit.c
  untracked:  commit.h
```

3 B) cat .pes/index output

```
gagangowda@DESKTOP-MBRTPOS:~/PES2UG24CS124-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776438143 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776438143 6 file2.txt
```

Phase 4

4 A) pes log output

```
./pes log
Initialized empty PES repository in .pes/
Committed: a5f64348f3ea... Initial commit
Committed: 8dd1ea26c1bf... Add world
Committed: d95f394a2f2b... Add farewell
commit d95f394a2f2be9f3a2c288263113935c839b3301eee916bcb2377fff6e37abae
Author: C.K Gagan Gowda <PES2UG24CS124>
Date:   1776438783

    Add farewell

commit 8dd1ea26c1bfd92411e460092485265b59c3e553a63de514d05b57cd09f693fc
Author: C.K Gagan Gowda <PES2UG24CS124>
Date:   1776438782

    Add world

commit a5f64348f3ea623265110e7cbd606507e5eebfbbca64b5a26cfbddd11ec04565
Author: C.K Gagan Gowda <PES2UG24CS124>
Date:   1776438782

    Initial commit
```

4 B) `find .pes -type f | sort` showing object growth

```
gagangowda@DESKTOP-MBRTPOS:~/PES2UG24CS124-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/27/b374421936320a8c527323a292a8f9eb9badf9be809f6137b8d32709ba7217
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/3f/50050781db8965d5fafbdbbdb632546637625ba2058dedef5c7ec14aae3a6
.pes/objects/53/623c175c06551b7365604fa414c63bf74f9677a178c9f8e60612018b4e0edf
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/8d/d1ea26c1bfd92411e460092485265b59c3e553a63de514d05b57cd09f693fc
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/a5/f64348f3ea623265110e7cbd606507e5eebfbbca64b5a26cfbddd11ec04565
.pes/objects/c9/6f2fc25394979be592d5cf80c5d78ec792ff13dde17f838449fbcdfbb7df0d
.pes/objects/d9/5f394a2f2be9f3a2c288263113935c839b3301eee916bcb2377fff6e37abae
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/refs/heads/main
```

4 C) `cat .pes/refs/heads/main` and `cat .pes/HEAD`

```
gagangowda@DESKTOP-MBRTPOS:~/PES2UG24CS124-pes-vcs$ cat .pes/refs/heads/main
d95f394a2f2be9f3a2c288263113935c839b3301eee916bcb2377fff6e37abae
```

```
gagangowda@DESKTOP-MBRTPOS:~/PES2UG24CS124-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

Final Full integration test

```
gagangowda@DESKTOP-MBRTPOS:~/PES2UG24CS124-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)
```

```
--- First Commit ---
Committed: 24bd3c98ae29... Initial commit

Log after first commit:
commit 24bd3c98ae29f22b267e4b2719aadc2d3bfb46a40db11514ba4fd10769ff4987
Author: C.K Gagan Gowda <PES2UG24CS124>
Date:   1776439101

    Initial commit


--- Second Commit ---
Committed: b596be512c83... Update file.txt

--- Third Commit ---
Committed: 6517801db05b... Add farewell

--- Full History ---
commit 6517801db05bd565da53c13744ebf27176f38994dd4d0e26e801d49d0d2e356f
Author: C.K Gagan Gowda <PES2UG24CS124>
Date:   1776439102

    Add farewell
```

```
commit b596be512c837ce31a31a63d48948fbf4f89f745613ece39140a6dc6fb3c8ef5
Author: C.K Gagan Gowda <PES2UG24CS124>
Date:   1776439102

    Update file.txt

commit 24bd3c98ae29f22b267e4b2719aadc2d3bfb46a40db11514ba4fd10769ff4987
Author: C.K Gagan Gowda <PES2UG24CS124>
Date:   1776439101

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
6517801db05bd565da53c13744ebf27176f38994dd4d0e26e801d49d0d2e356f
```

```
--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/24/bd3c98ae29f22b267e4b2719aadc2d3bfb46a40db11514ba4fd10769ff4987
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/65/17801db05bd565da53c13744ebf27176f38994dd4d0e26e801d49d0d2e356f
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/b5/96be512c837ce31a31a63d48948fbf4f89f745613ece39140a6dc6fb3c8ef5
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
```

Phase 5 and 6

Analysis question answers

Q 5.1

Update .pes pointers by verifying the branch file exists at .pes/refs/heads/, reading the target commit hash, and rewriting .pes/HEAD to ref: refs/heads/ using an atomic temp + rename. Resolve the target snapshot by reading the commit object, its root tree, and recursively expanding tree objects to build the full path → blob mapping.

Update the working directory by writing each blob to its file with correct mode, removing tracked files not present in the target tree, and recreating directories as needed. Rebuild the index by setting .pes/index entries to match the target tree with updated stat metadata.

This is complex because it must safely handle deletions, nested directories, permissions, partial failures, and avoid data loss when uncommitted changes overlap with target changes.

Q 5.2

Use a 3-way comparison per tracked file with base as the current HEAD tree blob hash, staged intent as the hash in .pes/index, and working file as stat plus recomputed blob hash. Build the target tree mapping and for each path differing between current HEAD and target, check if the working directory is dirty relative to the current committed state.

If the file is dirty and the target also changes that path, refuse checkout. Also detect untracked file overwrite cases where an untracked working file would be overwritten by the target tree and refuse.

This enforces the rule that checkout should not overwrite local non-committed content.

Q 5.3

In detached HEAD, .pes/HEAD stores a commit hash directly instead of a branch ref, and new commits still work by pointing to the previous detached commit while HEAD moves forward. The risk is that no branch references these commits, making them potentially unreachable.

Recovery involves creating a branch ref pointing to that commit chain or switching to a branch after anchoring the detached commit. If not recovered and no refs point to them, future GC may delete them.

Q 6.1

Use a mark-and-sweep approach where the mark phase initializes a hash set, seeds it with all branch tips, and performs DFS or BFS traversal marking commits, trees, and blobs while enqueueing referenced objects. The sweep phase scans all files in .pes/objects and deletes those whose hashes are not in the reachable set, optionally cleaning empty directories.

A hash set ensures O(1) expected membership checks and deduplication during traversal. In scale, traversal visits unique reachable objects, typically hundreds of thousands to a few million, making the complexity proportional to reachable objects plus total objects on disk.

Q 6.2

A race occurs when commit writes blob and tree objects, then commit object, and finally updates the branch ref while GC runs before the ref update and cannot see the new chain. GC treats new objects as unreachable and deletes them, after which the branch points to a broken history with missing objects.

Real systems avoid this through synchronization and locking around ref updates and GC, reachability protections like reflogs and grace periods, avoiding immediate pruning of recent objects, and ensuring objects are durable before references move.